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A Mathematical Formulation of AGI in the (C, U, V) Framework

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ABSTRACT

This article seeks to explore a general formulation of artificial general intelligence (AGI) under a unified framework that defines AGI agents as points in a joint (C, U, V) space. An agent is characterized by three components: ① a cognitive architecture C , which represents the modules (mathematical functions) inside the agent's mind, as well as the connections and communication protocols between these modules, including the theory of mind (ToM); ② a set of potential functions U , which represents the skills of perception, cognition, and planning (e.g., a potential function can be a neural network trained for visual object recognition, or embodied motion planning); and ③ a set of value functions V , which includes the agent's urges, preferences, and social affections, as well as benefits for individual agents or a group of agents. In this setting, "intelligence" is defined as a wide range of phenomena exhibited by agents when they interact with complex environments (i.e., physical intelligence) and other agents (i.e., social intelligence). Given an initial point in the (C, U, V) space, an agent can explore new V -dimensions, which in turn drives the acquisition and learning of skills by enabling the learning of new potential functions U in the environment and by updating the cognitive model. We have developed a Tong test as a benchmark and evaluation criteria: An agent that has reached the human level (C, U, V) is called a "Tong Agent." The convergence of this process defines the limits of the agent's evolution; we name this the "stopping problem" of Tong Agents, based on the analogy of the halting problem in a Turing machine.

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1. Introduction

In this paper, we seek to provide a definition and a learning framework for artificial general intelligence (AGI). To define AGI, it is crucial to understand how an intelligent agent perceives and interprets its environment. At the core of this understanding lies the concept of the parse graph (PG). For an intelligent agent, a scene is typically interpreted as a collection of interrelated concepts and functions. These concepts and functions are further utilized by the agent to achieve its objectives. For example, when an agent observes a kitchen scene, it may interpret it as a PG in which nodes represent objects such as a stove, a pot, and ingredients,

while edges capture relationships such as "the pot is on the stove" or "the stove heats the pot." In a social scene, a PG can further represent the relationships, roles, and interactions between entities. Each node in the graph corresponds to an individual, object, or abstract concept, while edges capture the hierarchical or relational dependencies among them. Different species may have distinct PGs, based on their living environments and levels of intelligence. Even within the same species, such as humans, individuals may construct different PGs based on their unique experiences and perspectives.

We believe that modeling the construction and evolution of PGs is essential for understanding intelligent agents. To this end, we have developed a mathematical framework called the cognition-potential-value framework. Fig. 1 illustrates the proposed framework characterized by a cognitive architecture C , a set of potential functions U , and a set of value functions V , which underpins our theoretical formulation of AGI. It visually summarizes the

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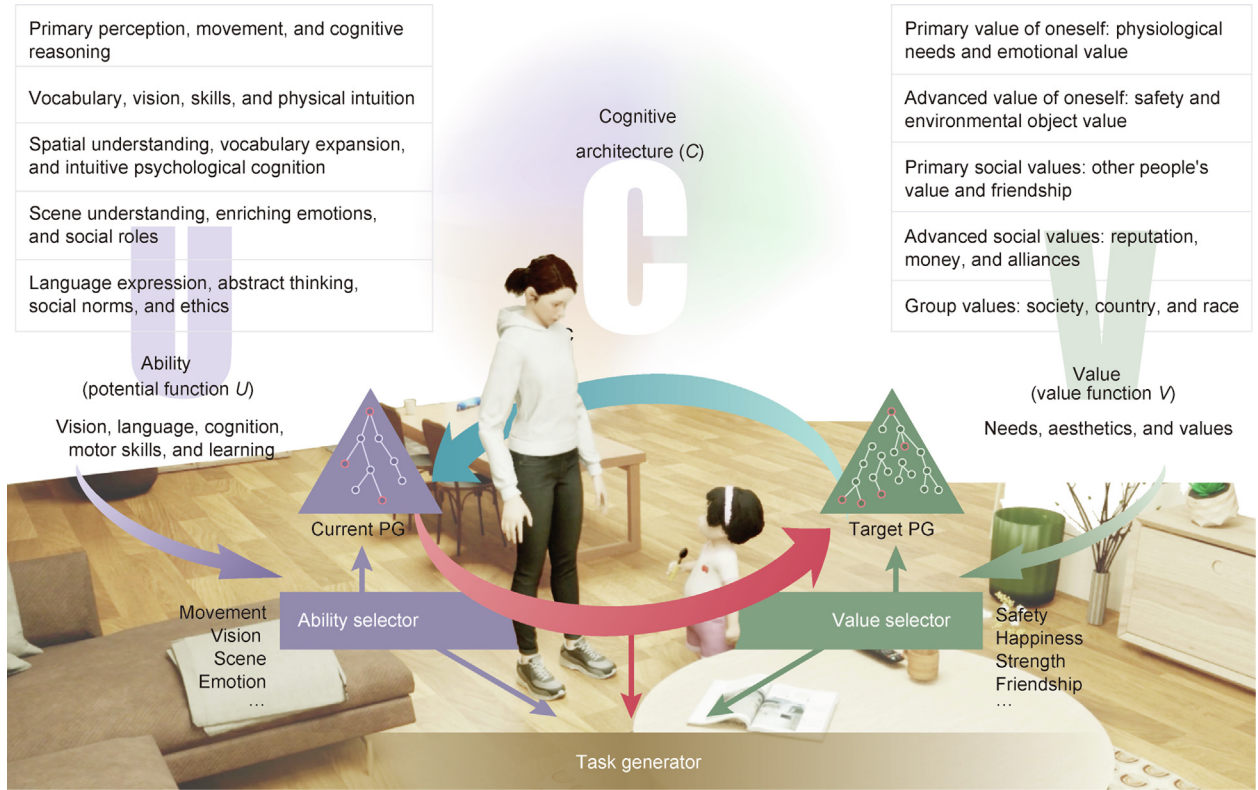


Fig. 1. The (C, U, V) framework for AGI: The cognitive architecture selects skills to generate tasks that increase value dimensions, forming a co-evolving learning loop.

interactions between the cognitive architecture, the skill space, and the value system, giving an intuitive overview of how these elements co-evolve. This framework provides a systematic approach to describe how agents interact with their environments, acquire knowledge, and develop skills to increase their intelligence. “ U ” represents the set of potential functions, which reflects the skill sets of an intelligent agent, encapsulating its ability to perceive, understand, and interact with its environment and other agents. These skills are foundational to the agent’s intelligence and adaptability, encompassing both basic perception and advanced reasoning capabilities. For example, basic skills such as image classification and object recognition can be modeled using neural networks, decision trees, or support vector machines (SVMs) [1–5]. These capabilities enable an agent to identify and categorize objects within a scene. More advanced skills—such as understanding causal relationships, interpreting abstract concepts, or predicting human intentions—may require the use of sophisticated, parameterized models. The parameters of these models can be optimized using various learning paradigms. “ U ” also reflects the agent’s understanding of the environment and its interactions with other agents. For instance, when observing a kitchen scene, an agent with poor skills might only identify objects such as a stove and a pot, while a more skilled agent would recognize their causal relationship—for example, that the stove heats the pot. Similarly, in social scenarios, “ U ” enables the agent to understand roles, interactions, and relational dynamics, such as predicting group behavior or identifying hierarchical structures. The skill sets evolve with the agent’s experiences and goals, improving as the agent encounters new tasks and challenges. By encapsulating both practical skills and conceptual understanding, “ U ” serves as the foundation for the agent’s ability to act effectively and intelligently across diverse scenarios, bridging perception and action in meaningful ways.

“ V ” represents the value functions guiding an agent’s decisions and actions; it evolves through experiences and expands as the

agent’s knowledge grows. The V system can be considered a high-dimensional vector, with each dimension corresponding to a specific human need, such as health, wealth, and peace. Starting with self-oriented priorities, such as safety and comfort, agents first fulfill higher values within these basic dimensions before developing values in more advanced dimensions through social interactions, such as responsibility and, ultimately, collective well-being. The evolved value dimensions can be learned from demonstrations by experienced agents or transferred autonomously. This hierarchical structure mirrors the progression from individual needs to broader societal values. The dynamic interplay between U and V enables the agent to adapt to diverse environments, solve problems, and exhibit intelligent behavior. The varying (C, U, V) structures across agents or species reflect the diversity in how they perceive, prioritize, and engage with the world.

The cognitive architecture C learns to select skills from U , choose value dimensions from V , and generate tasks and actions aimed at increasing the values in the selected dimensions. In general, the cognitive architecture consists of several key modules. One such module is the value selector, which defines the agent’s current objective. For example, during infancy, a person may prioritize basic needs such as nourishment and comfort, focusing on actions that satisfy those values. In contrast, during youth, the goals may shift to learning, social interaction, or skill acquisition, with the value selector emphasizing dimensions such as knowledge, relationships, or personal growth. This adaptability allows the cognitive architecture to align its actions with evolving goals over time. Correspondingly, a skill selector is employed to identify potentially useful skills that can interpret the scene and further guide the agent in taking appropriate actions to increase the selected values. Based on the selected skills, the agent uses a PG generator to construct the PG of the current scene. Furthermore, it generates target (future) PGs, representing the desired states that the agent expects will increase the selected value once achieved. The trajectory from the current PG to the target PG

defines a specific task, and the agent employs an action policy to interact with the scene. This policy is designed to maximize the task reward, thereby correspondingly increasing the selected values.

We define AGI as the interplay among the above three elements: the cognitive architecture, the potential (skill) functions, and the value functions. These elements are in a state of co-evolution to describe the formation of AGI, where learning and evolution are not isolated but are intricately linked within this space. At any given time step, the cognitive model can choose value dimensions in V and skills in U . The agent will then continuously learn and adapt new skills, and U will evolve. The cognitive architecture builds the PGs and develops reasonable tasks to maximize its values. As the agent acts and the values in the chosen dimensions are maximized, V evolves, expanding the agent's value space and allowing for a deeper understanding of its objectives. As the agent interacts with its environment and learns more, its cognitive model also evolves, enabling the agent to select more appropriate skills and design more effective tasks to increase its values. This evolution strengthens the cognitive architecture, similar to how the human brain develops and grows more sophisticated over time, leading to enhanced abilities and intelligence. In essence, the co-evolution of the (C, U, V) framework is a cyclical process in which each element influences and is influenced by the others. The cognitive model shapes how the skill functions are learned and applied, which in turn affects how the value functions are perceived and maximized. This cycle of learning, interaction, and evolution is what distinguishes AGI, and the experience of any intelligent agent can be expressed as a trajectory in this (C, U, V) space.

The (C, U, V) framework can unify most of the existing learning methods described in the literature. For example, supervised learning tasks such as image or sentence classification can be situated within the (C, U, V) framework as a "skill" in U , where the learner receives signals passively and aims to improve a specific type of accuracy. Active learning tasks can be situated in a similar manner, with the key difference being that the learner actively engages to improve the corresponding skill. Reinforcement learning (RL) can be placed in C , where the learner attempts to maximize a specified task through carefully designed rewards. However, none of these approaches, whether passive or active, can fully define AGI, as human intelligence is far more complex and adaptable than the types of learning processes captured by these frameworks. We believe that the (C, U, V) framework, which is useful for understanding structured learning systems, can account for the depth and flexibility of human cognition, which encompasses not only learning from experience but also the ability to reflect, innovate, and make complex decisions with emotional, social, and ethical considerations. Based on the (C, U, V) framework, we propose fundamental theoretical questions about the convergence and limits of (C, U, V) and define the concepts of cognitive and task completeness. We also discuss open problems in response to our view of AGI in the discussion section.

2. Defining the agent's skills by the U space

The U space represents the skill set of an intelligent agent, where each skill is tailored to certain specific tasks. This skill set encompasses a range of capabilities, from visual tasks such as object recognition and motion planning to linguistic skills such as name entity recognition and sentiment classification, as well as motor skills, cognition, and learning. Each skill within the U space is considered a basic function with a particular role, and these functions are parameterized, allowing them to be learned and refined through various learning methods. The learning

processes are not restricted to a single paradigm but are arbitrary, including but not limited to self-supervised learning [3], where the agent learns from provided examples and unlimited data; active learning, where the agent seeks out information itself; and generally communicative learning, where agents gain knowledge through interaction and communication [6,7].

The flexibility of the U space allows the agent to adapt and enhance its skills over time, which is crucial for AGI. As the agent learns and interacts with its environment, its skills in the U space evolve, enabling it to perform tasks more effectively and exhibit intelligent behavior across a spectrum of scenarios. This dynamic nature of the U space is integral to the development of an agent's overall intelligence and its ability to engage with difficulty. The U space is composed of a set of potential skills:

$$U = \{u_1, u_2, \dots, u_N\} \quad (1)$$

We denote x as any scene that contains observable information, such as visual or linguistic data. Each skill u_i ($1 \leq i \leq N$) takes x as input and produces skill-dependent signals as output. The accuracy of each skill u_i can vary among different agents, reflecting their unique capabilities and experiences. We define the activated U space of the current agent as those skills with non-trivial accuracy; that is, only those skills that perform above a certain threshold are considered part of the agent's activated skill set. This subset of skills is crucial for the agent's functionality. While the total number of potential skills N can be extremely large, the activated U space (i.e., the number of activated skills with non-trivial accuracy) can be relatively small, highlighting the selectivity in skill development. The time-evolved U space is represented as follows:

$$U_t = \{u_{1,\theta_{1,t}}, u_{2,\theta_{2,t}}, \dots, u_{N,\theta_{N,t}}\} \quad (2)$$

where each $u_{i,\theta_{i,t}}$ denotes a skill u_i with its respective parameter set (e.g., neural network weights) $\theta_{i,t}$ at time t . For example, a skill can be modeled by a deep neural network, where the set of parameters can be trained by gradient descent algorithms. The parameters are pivotal in determining the proficiency of each skill and directly influence the agent's capability to perform various tasks. As the agent engages in learning and adaptation, the parameters are updated to give better accuracy, which in turn leads to the evolution of the activated U space. The iteration of parameter updates and skill advancements is fundamental to the expansion and improvement of the activated U space.

The U space not only evolves over time but also exhibits a hierarchical structure that reflects the progressive development of an agent's skills. At the foundational cognitive stage, humans acquire practical skills such as movement, visual perception [3], language comprehension [2], and basic knowledge. This initial level of the U space is designated as U_0 , where individuals learn essential skills to meet their immediate needs and construct a rudimentary representation of the world. In terms of the PG, this stage can be denoted as PG_0 . As individuals progress and find that their basic skills sufficiently address their primary needs, they begin to develop more advanced skills for interacting with others. This marks the transition to a higher level of the U space, which we denote as U_+ . The corresponding PG at this level is PG_+ , and it includes additional representations of other individuals, reflecting the development of social skills such as the ability to estimate or predict others' intentions, motivations, and emotions. Further along the hierarchy, when humans embrace collective concepts such as family, community, and nation, they naturally require more sophisticated skills in social communication. This level of the U space is referred to as U_{++} . The PG at this stage is denoted as PG_{++} , incorporating even higher-level information and representing the agent's more complex social interactions and collective understanding. An agent's U space is dynamic and can evolve and transition to higher levels.

As the agent's lower-level needs are satisfied and it starts to focus on higher-level needs, its U space will transition into a more complex space, prompting the development and learning of advanced skills. Consequently, its PG will incorporate higher-level information, reflecting the agent's growing cognitive and social capabilities. This hierarchical and evolutionary nature of the U space is integral to the agent's ability to adapt and engage with increasingly complex environments and tasks.

3. Driving the agent's actions by the V space

For an intelligent agent, we define the value space, V space, as the system governing its values, preferences, and guiding principles for decision-making. Based on its current state and existing knowledge encoded in a function U , the agent selects actions that align with its intrinsic value system. This mechanism drives the agent from interpreting the objective world to making subjective choices, encapsulating its unique value structure. Each agent possesses distinct values, exhibits preferences for different strategies, and acts in ways that prioritize fulfilling its value dimensions. These dimensions, such as health, wealth, peace, and social responsibility, collectively form the agent's V space.

The concept of V space can be applied to understand differences across species and contexts. For many animals, survival primarily revolves around basic physiological needs and safety, corresponding to a simpler V space. For instance, animals perform tasks such as hunting, mating, and avoiding danger in order to sustain life. Through these activities, they develop familiarity with their environments, acquire relevant skills, and establish their knowledge space, or U space. The relationship between the V space and the U space highlights the interplay between values and knowledge in shaping intelligent behavior. While different species may share similar V space dimensions, such as survival needs, their U spaces can vary significantly due to differences in living environments and physical structures. For example, both terrestrial creatures and aquatic species prioritize survival, but the knowledge they acquire and the natural laws they navigate differ substantially. These differences underscore the diversity in how agents perceive, prioritize, and engage with their environments.

Building on this concept, we further argue that the V space is dynamic and evolves with the agent's experience and knowledge. Numerous facts demonstrate that the value space often has hierarchical levels, with different agents possessing value spaces with varying levels and scopes. For example, the ancient Chinese saying, "The superior man is aware of righteousness, the inferior man is aware of advantage," illustrates how individual value spaces can differ. As an agent acquires more knowledge and encounters more diverse scenarios, its value function and preferences may change and expand accordingly. Consequently, the agent might make different decisions in similar scenarios based on its evolved value space; moreover, the agent's value space will expand, leading to the emergence of more high-level values and new tasks.

When humans are at the elementary cognitive stage, the values they consider are mostly based on individual interests; for example, they may prioritize physiological needs such as health, safety, and satiety, along with personal emotional needs such as pleasure, comfort, and cleanliness, in order to maximize individual benefits. We refer to this level of value space as V_0 , representing the primary value of maximizing individual interests. Once individual interests are sufficiently met, humans begin to consider interaction between individuals. At this stage, more attention is given to values related to recognition, privacy, and belonging, as well as values related to others, such as reputation, etiquette, and responsibility. These values can be collectively referred to as the values at the level of individual-to-individual interaction; they are denoted as V_+ . Once

the needs related to others are fulfilled, the individual further considers collective and social values, such as those related to societal governance, including democracy, civilization, and the rule of law. These values are collectively referred to as values at the level of the individual and the group, or the community of interest, and are denoted as V_{++} . Furthermore, value prioritization can vary significantly across agents due to their individual experiences or cultural backgrounds. In our framework, this is modeled by agent-specific value preference parameters, which serve as a latent embedding encoding individual- or group-level biases. For example, one agent may prioritize "wealth" over "health," while another may reverse this order, depending on its contextual upbringing, social exposure, or cultural values. These differences influence the output of the value selector and shape the agent's behavior over time. More discussion on this topic can be found in [Section 4.1](#).

The evolution (or expansion) of the V space can be driven intrinsically or through demonstrations by experienced agents. For example, as children, we expand our value systems by imitating our parents, learning from their behaviors and guidance. This hierarchical and dynamic evolution of the V space also facilitates the agent's ability to solve diverse problems, adapt to complex environments, and exhibit intelligent behavior. The interplay between knowledge and values underscores the adaptability and diversity of intelligent agents. By analyzing the structure and dynamics of the V space, we can better understand how agents make decisions, prioritize actions, and navigate the complexities of the real world.

4. Cognitive architecture

The goal of AGI is to develop computational agents that exhibit human-level intelligence. One key distinction between humans and other animals is the human ability to learn, select, and apply various skills to tackle complex tasks in order to meet hierarchical needs. This distinction may be summarized as a term called "cognitive architecture." For example, humans can use their hands to grasp tools and use language to communicate abstract concepts, plan for the future, and collaborate with others, enabling them to accomplish goals that require coordination and the integration of diverse skills. Their cognitive architecture—which not only enables humans to learn from experience but also allows them to adapt strategies based on changing circumstances—sets humans apart from other species. In striving to replicate this capacity, the developers of AGI aim to create systems capable of flexible reasoning, decision-making, and continual learning, similar to how humans navigate the complexities of daily life. Importantly, the cognitive architecture is not a single algorithm that helps artificial intelligence (AI) algorithms solve a specific problem; rather, it serves general-purpose intelligence and represents the process by which AGI learns, encodes, and applies knowledge independently of a specific task to produce behavior [8,9]. Through detailed analogies to the composition of human cognition, developmental functions, and mechanisms, the cognitive architecture of an AGI enables the expression of human-like intelligent behaviors in diverse environments. Past cognitive architectures include Soar [10], adaptive control of thought-rational (ACT-R) [11], and CLARION [12]. However, current cognitive architectures still have multiple limitations, such as being based on different assumptions and theories and therefore lacking comparability, being only able to handle specific simple tasks, encountering challenges in generalization, and lacking inspiration from biological mechanisms.

Therefore, in addition to the U and V spaces, we introduce the cognitive architecture, which describes how agents select specific skills and generate tasks to accomplish their needs, thereby increasing their values. The cognitive architecture consists of the following modules, each of which serves a specific function: the

value-selection module, the skill-selection module, and the cognitive core, which includes the cognitive understanding module and the cognitive planning module. These modules are discussed in more detail below.

4.1. Value-selection module

Given a scene x , the value-selection module determines the dimensions of value that the agent prioritizes during the current period, based on the agent's characterization and preferences. For example, consider a scenario in a kitchen where the agent is hungry. In this case, the agent may focus on value dimensions related to sustenance, giving these values priority over other potential needs. These values could reflect immediate requirements, such as food, as well as goals such as health, satisfaction, and well-being, guiding the agent to develop corresponding tasks and take appropriate actions. It is important to note that value selection should be personalized. An extremely diligent agent, for instance, may prioritize completing work over comfort, thus choosing the dimension related to responsibility over that linked to comfort. Given the profile of the agent denoted by θ_{agent} , we define the value-selection module as

$$V_{\text{select}}(V; \theta_{\text{agent}}) \quad (3)$$

which takes as input the set of values and the agent's profile θ_{agent} , and outputs a set of dimensions in V . V_{select} represents the selected values. This output indicates the selected value dimensions at the present time period, with each element indicating whether a particular value dimension has been activated. θ_{agent} personalizes the value preference, allowing the module to prioritize values based on different agents' personalization, and can be interpreted as encoding both innate preferences and learned cultural or societal norms. These parameters allow the agent to exhibit context-aware trade-offs. For instance, in situations involving a choice between health and wealth, θ_{agent} determines which value dimension is prioritized. This enables the modeling of cultural variation—such as individualism versus collectivism—within the same high-dimensional value representation. Once the value dimensions are selected, the cognitive architecture learns to select skills, adjust the agent's behavior, and generate tasks accordingly, ensuring that the tasks and actions align with V_{select} and contribute to the agent's interests.

4.2. Skill-selection module

Given an agent with skills $u_1, u_2, \dots, u_N$, the skill-selector module chooses the skills, driven by the agent's current needs V_{select} . For example, in the same scenario described above, where the agent is hungry, the agent may choose the set of skills appropriate to address its need for sustenance. The skill-selector model might thus activate the following skills:

- A visual recognition skill, which allows the agent to scan the kitchen environment, identify food items, and determine which items are suitable for consumption; the agent may use this skill to locate ingredients or prepared food.
- A language skill, which helps the agent gather additional information, issue decision-based instructions, or communicate with others (e.g., a virtual assistant or a human); the agent could use this skill to read a recipe or ask for clarification on the task.
- A motion skill, which helps the agent navigate the environment based on the information gathered from the visual recognition and language skills; the motion skill enables the agent to physically interact with objects in the kitchen, such as reaching for food, opening the refrigerator, or cooking.

By dynamically leveraging and combining these skills, the agent can efficiently take actions to satisfy hunger. In this case, the skill-selector model learns to choose the right combination of skills given the situation, enabling it to increase the values in the dimensions. The process also involves continual adjustment based on the agent's environment and evolving needs.

Formally, we define the skill-selection module as

$$U_{\text{select}}(U, V_{\text{select}}, x; \theta_{U_{\text{select}}}) \quad (4)$$

which takes as input the set of skills, the scene x , and the selected values V_{select} , and outputs a set of skills, indicating the selected skills to improve the selected value dimensions. U_{select} is the selected skills. The learnable set of parameters $\theta_{U_{\text{select}}}$ governs the selection process. Once the skills are selected, the cognitive architecture uses them to develop tasks to address the agent's needs.

4.3. Cognitive core

Given the selected value dimensions V_{select} and skills U_{select} , the cognitive core module understands the scenes with the skills and generates tasks that can increase the values in the dimensions. To achieve this, the cognitive core module contains a cognitive understanding module and a cognitive planning module.

4.3.1. Cognitive understanding module

The cognitive understanding module constructs a representation of the scene by utilizing the selected skills U_{select} . As discussed above, humans typically parse any given scene into a PG. Similarly, the cognitive understanding module leverages the outputs of various skills to generate an analogous PG, which encapsulates the key components and relationships of the scene x .

For instance, in the context of a kitchen scene, if the agent selects skills such as visual recognition and motion, the PG generated for the kitchen will consist of various objects and their spatial locations. It will also capture the interactions between the agent and these objects, detailing actions such as picking up a cup, opening a cabinet, or placing ingredients on the counter. In this way, the PG not only models the static entities present in the scene but also incorporates dynamic interactions, enabling a richer understanding of the environment and the agent's activities within it. This approach allows the cognitive understanding module to create a comprehensive and contextually accurate representation of the scene—one that mirrors how humans cognitively process and interpret their surroundings in real-world scenarios. Mathematically, we denote the generation of the PG as

$$\text{PG}_{\text{Gen}}(x, U_{\text{select}}; \theta_{\text{PG}}) \quad (5)$$

which takes the scene x and selected skills U_{select} as input and outputs a PG to describe the scene. PG_{Gen} can be any complex function class, such as a neural network. For practical implementation, the model parameter θ_{PG} can be instantiated as a learned graph neural network combined with a neural-network-based object detector. The object detector can be trained to identify objects and their spatial locations within an image. Given these detected objects and their locations, a graph neural network can then construct the corresponding PG by processing these signals as input. Both components are trained on large-scale labeled datasets; through this training, the models can generalize to unseen scenarios, thereby avoiding the risk of a combinatorial explosion of graph construction. It is important to note that different agents may perceive the same scene differently, and the ability to generate all possible PGs is crucial for accurately describing any agent's understanding of the whole population. We define the capacity of PG_{Gen} to express all possible PGs as "cognitive completeness," which will be further discussed in the next section.

4.3.2. Cognitive planning module

With the PG representing the current scene, the agent aims to increase its values in V_{select} by performing a series of actions. See Fig. 2 for a concrete demonstration. In human cognition, abstract tasks are often developed first. For instance, consider a hungry agent in a kitchen. The agent can devise a range of tasks, each of which contributes to increasing the value in the sustenance dimension. These tasks could include cooking food from scratch or simply heating pre-prepared food in the microwave. Each of these actions reflects a different approach to achieving the same goal—satisfying the agent's hunger. Interestingly, the completion of these tasks can also be represented as a PG, akin to how the goal scene is described. In this future PG, the agent's actions and their resulting changes in the environment are mapped out, illustrating how each step contributes to the overall task completion. This recursive structure—in which the agent's actions unfold within a PG and lead to the formation of new PGs for task progression—offers a dynamic and layered understanding of how agents interact with their environments in a goal-directed manner. It is clear that the planning module should depend on the selected values; we denote the future PG ($\text{PG}_{\text{PlanGen}}$) that the agent imagines can increase the value as

$$\text{PG}_{\text{PlanGen}}(x, U_{\text{select}}, V_{\text{select}}; \theta_{\text{PG}_{\text{plan}}}) \quad (6)$$

Similar to PG_{Gen} , $\text{PG}_{\text{PlanGen}}$ with its parameter set $\theta_{\text{PG}_{\text{plan}}}$ can be implemented by a neural network. Correspondingly, we can define the task completion as finding all possible future PGs to increase the value and finding a path from the current PG to the future PGs.

We define this capacity as “task completeness.” Once the task is defined and the associated rewards are specified, any learning algorithm—whether supervised learning, active learning, or RL—can be employed to take appropriate action. It is highly likely that the agent will only be able to find an approximated solution. Let us consider a kitchen scenario in which the agent aims to prepare a delicious meal. However, due to the agent's limited cooking skills and lack of experience, the result may not meet expectations; that is, the dish may fall short in taste, texture, or presentation. This outcome, while imperfect, is an essential part of the learning process.

It provides valuable feedback that enables the agent to refine and improve its skills over time. This process is much like how chefs improve their craft: through continuous practice, learning from mistakes, and gradually improving their abilities.

The value outcome in V_{select} is defined as an unknown underlying function of a PG. In the whole learning process, the agent first defines its V_{select} , selects the necessary skills, plans the required actions, and then executes them to obtain a result. This result, whether successful or not, provides critical feedback that is reflected in the values in V_{select} . By interacting with the world and observing the consequences of its actions, the agent gradually gains more experience and learns to select more appropriate actions. Over time, it can improve its planning process, optimize the skills it uses, and ultimately maximize the values represented in V_{select} . Just as a chef learns from each prepared dish, adjusting the ingredients, techniques, and timing based on feedback, the agent learns to adjust its strategies, refining its approach to achieve better outcomes in future attempts. Through this iterative process, the agent increases its intelligence and decision-making capabilities and moves closer to a higher-value space.

5. Learning in the (C, U, V) space

The question of how to define an intelligent agent has been pursued persistently throughout human history by a wide range of religious scholars, philosophers, and artists [13,14]. However, it was not until cognitive scientists and psychologists began to define intelligence as a measurable concept that it became an object of scientific inquiry [15–18], exemplified by the development of the Binet–Simon intelligence test [19]—the world's first intelligence test. In the realm of intelligence research, particularly concerning human intelligence, researchers have proposed various theories and frameworks to explore the components of human intelligence, such as Howard Gardner's theory of multiple intelligences [20] and Daniel Goleman's theory of emotional intelligence [21]. Despite extensive academic efforts, there remains a lack of consensus on what “intelligence” truly entails, and a unified theory has yet to be established [22–24].

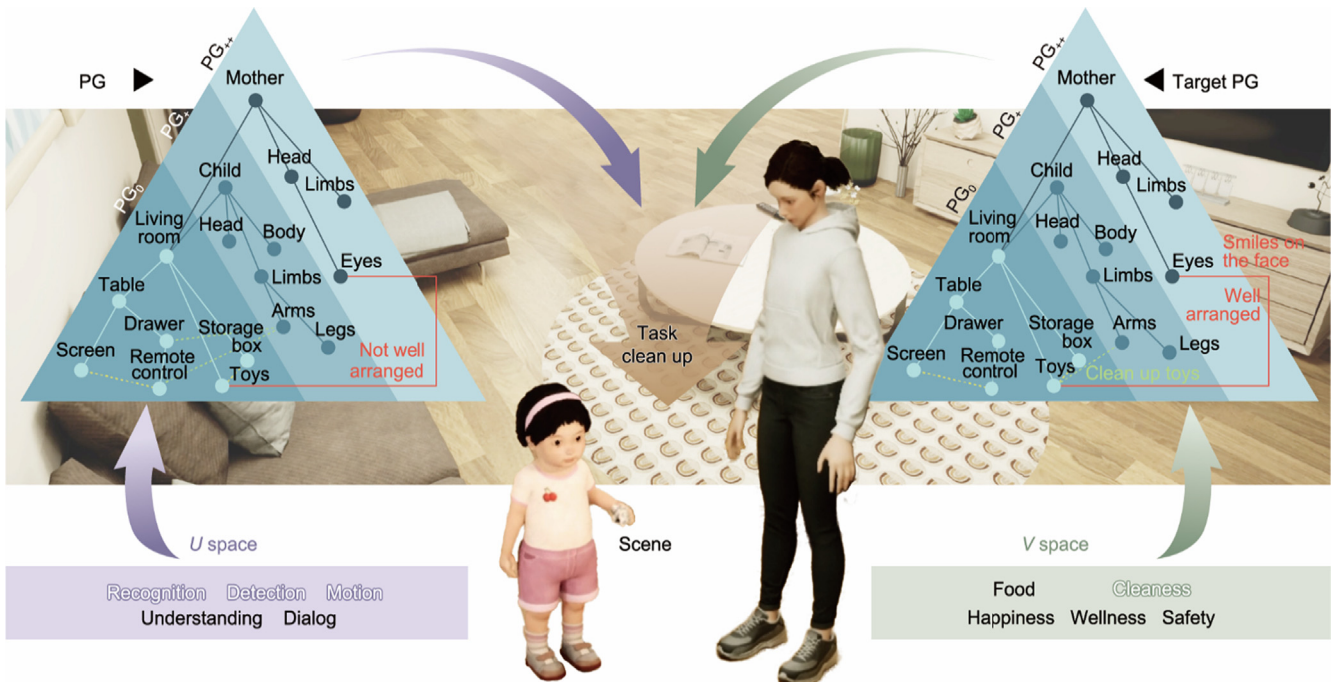


Fig. 2. A concrete example: Using cleanness as the value, a child generates PGs using related skills and develops tasks (clean up the room) to increase the value (cleanness).

We define AGI as the learning process in the (C, U, V) space, which emphasizes how agents adapt and refine their understanding, skills, and objectives through interaction with their environment, the experience of any intelligent agent can be expressed as a trajectory in this (C, U, V) space, as illustrated in Fig. 3. Learning challenges in this space are multifaceted, spanning environmental representation, decision-making, feedback interpretation, and long-term adaptation. This section elaborates on the nuanced problems that arise in the (C, U, V) space and the strategies agents employ to address them. We formalize this learning problem as an online process, which can be abstractly described in Algorithmic 1.

Algorithm 1 (Online) Learning in the (C, U, V) framework.

Input: Parameterized C_0 , U_0 , and V_0 at time $t = 0$

Output: C_T , U_T , and V_T at a given time T

For $t = 1$ to T :

1. **Get V_{select} :** Use V_{select} to prioritize value dimensions based on the agent's current state and profile

2. **Get U_{select} :** Use U_{select} to identify the needed skills

3. **Generate PG:** Construct the current scene's PG using PG_{Gen}

4. **Plan task:** Define the target (future) PG that maximizes the selected values and determine the required actions to transition from the current PG to the future PG using PG_{PlanGen}

5. **Interact with environment:** Perform actions that interact with the environment

6. **Receive task reward:** Collect feedback on task completion

7. **Receive value feedback:** Observe how the actions impacted the selected value dimensions.

8. **Update selectors:** Adjust the parameters based on feedback to improve future decisions

5.1. Learning in the UV system

The first step in the (C, U, V) learning framework is to identify the value dimensions that the agent should prioritize at a given moment. The agent employs a selector module to analyze its current state and profile, determining which value dimensions align most closely with its immediate and long-term goals. For instance, in a kitchen scenario, the agent might prioritize health and sustenance if its goal is to prepare a nutritious meal. The selector must take into account external factors, such as environmental constraints, and internal states, such as the agent's energy levels or predefined preferences. This prioritization requires the agent to weigh the trade-offs between competing values, ensuring that its objectives remain achievable and relevant to the situation.

Challenges arise in managing complex and dynamic value hierarchies. As an agent encounters new environments or gains experience, its value dimensions may evolve. For example, an agent initially focused on individual needs might shift toward collective or societal goals, such as collaboration or sustainability. The selector must adapt to these changes, balancing flexibility with consistency to maintain coherent decision-making. Furthermore, the agent must address potential conflicts between short-term priorities and long-term objectives, such as choosing between a quick but unhealthy snack versus a time-intensive, nutritious meal. Resolving these conflicts requires sophisticated mechanisms for evaluating trade-offs and predicting outcomes.

The final step in the (C, U, V) learning framework is updating the V -selector and U -selector, which govern the agent's ability to prioritize values and select appropriate skills. These selectors are the linchpins of the framework, enabling the agent to adapt its behavior to changing circumstances and objectives. By adjusting the parameters of these selectors, the agent can refine its decision-making process, ensuring that its actions remain aligned with its goals and values. Updating selectors involves integrating insights from task rewards and value feedback. For instance, if the agent consistently receives higher rewards for prioritizing health-related values, the V -selector might increase the weight

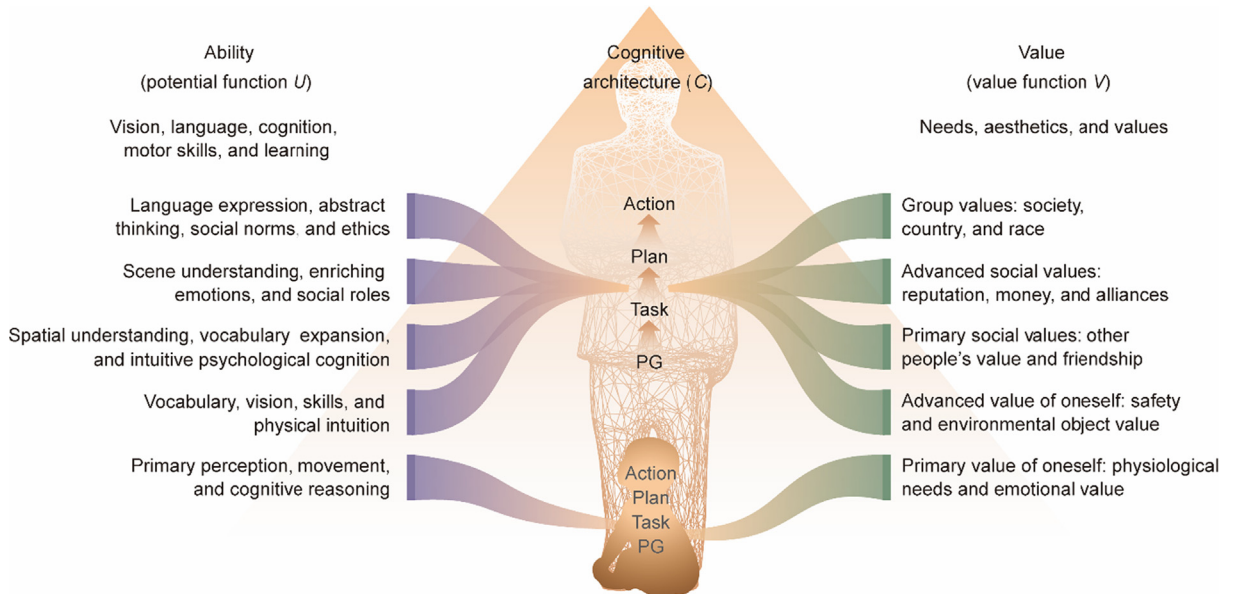


Fig. 3. Evolution of the (C, U, V) space corresponds to the growth of intelligence.

assigned to those dimensions. Similarly, if certain skills prove particularly effective in achieving specific objectives, the U -selector might prioritize those skills in future tasks. This feedback-driven refinement enables the agent to learn from its experiences, continuously improving its ability to navigate complex environments.

One of the key challenges in updating the selectors is balancing stability and flexibility. On the one hand, the selectors must remain consistent enough to provide reliable guidance and avoid erratic decision-making. On the other, they must be adaptable enough to respond to new information and changing circumstances. Achieving this balance requires sophisticated learning algorithms that can identify meaningful patterns and trends while filtering out noise and irrelevant signals. The selectors must also account for the hierarchical and multidimensional nature of the (C, U, V) space. For example, the V -selector might need to prioritize high-level goals, such as sustainability, while also addressing more immediate concerns, such as task efficiency. Similarly, the U -selector must navigate trade-offs between developing new skills and leveraging existing ones, ensuring that the agent's capabilities remain both versatile and specialized. By continuously refining these selectors, the agent can achieve a dynamic equilibrium that supports both short-term success and long-term growth.

5.2. Learning PGs

Once the value dimensions are selected, the agent identifies the specific skills required to achieve its objectives. The selector module analyzes the prioritized values and matches them with the agent's skill set. For example, if the agent's goal is to cook a meal, the selector might activate skills related to visual recognition, motor control, and recipe interpretation. This selection process ensures that the agent's capabilities are aligned with the demands of the task, thereby optimizing efficiency and effectiveness.

With the selected value dimension and skill sets, the agent leverages the graph-generation module to build a PG of the scene. The PG is a central component of the (C, U, V) learning framework, providing a structured representation of the agent's understanding of its environment. Constructing the PG involves identifying key entities, their attributes, and the relationships between them. Building an accurate PG presents several challenges: The agent must integrate sensory data from multiple modalities, such as vision and touch, to construct a coherent representation. This process requires resolving ambiguities, such as distinguishing between similar-looking objects, and handling incomplete or noisy data. Additionally, the PG must be dynamically updated as the environment changes, ensuring that it remains an accurate reflection of the current state. For example, if the agent slices a vegetable, the PG must add nodes for the sliced pieces and update the relationships between them.

5.2.1. PG learning strategy

Since PGs are general graphs, graph-generative models can be leveraged to establish the mapping from scenes to graphs. A wide range of function classes can be chosen from to build this generative model, such as graph neural networks. For parameter learning, a straightforward approach would be to apply supervised learning (i.e., learning from demonstrations). Since we have a dataset D consisting of scenes x and skills U_{select} from different agents, along with their corresponding PGs labeled by the agents, the model can be learned by solving the following optimization problem:

$$\min_{\theta_{\text{PG}}} \frac{1}{|D|} \sum_{(x, U_{\text{select}}, \text{PG}) \in D} \text{loss}(\text{PG}, \text{PG}_{\text{Gen}}(x, U_{\text{select}}; \theta_{\text{PG}})) \quad (7)$$

where the loss function can be implemented by a pairwise loss to measure the difference between two graphs.

Task planning involves defining a target state, represented by a future PG, that aligns with the agent's prioritized values. The agent must then identify the sequence of actions required to transition from the current PG to the target PG. This process requires anticipating potential obstacles and adjusting the plan accordingly. For effective task planning, the agent must conduct a path search of the PG and balance efficiency with adaptability. The plan must account for uncertainties, such as the availability of resources or unexpected environmental changes. Additionally, the agent must evaluate multiple potential strategies, selecting the one that maximizes the likelihood of success while minimizing resource use.

5.2.2. Challenges

PGs are representations of environments. The agent must interact with the environment by executing actions. For instance, if the agent plans to prepare a meal, it might execute actions such as chopping vegetables, boiling water, or setting the table. Each action modifies the environment, and the PG will update accordingly. The challenges encountered in this step stem from the dynamic and often unpredictable nature of real-world environments. Environments are rarely static; objects may move, external factors may intervene, and conditions may change unexpectedly, leading to an unpredicted PG. For example, the agent might discover mid-task that an ingredient is missing or a tool is malfunctioning. In such scenarios, the agent must adapt its plan in real time, recalibrating its strategy to accommodate these changes. Another critical aspect of this step is precision in execution. Actions must be carried out with sufficient accuracy to achieve the desired outcome. Errors in execution can lead to suboptimal outcomes or even task failure, prompting the agent to refine its motor skills and adjust its approach in subsequent iterations. The interplay between execution and observation also creates opportunities for real-time learning. As the agent performs actions and observes their effects, it gathers new information about the environment, which can be used to update its PG and refine its understanding. For instance, if the agent notices that boiling water takes longer than expected, it can adjust its timing for similar tasks in the future. This iterative loop of action and observation is fundamental to the agent's ability to learn and adapt effectively.

5.3. Learning to complete tasks by interacting with the environment

Once the agent interacts with its environment and completes a task or action, it receives task-specific feedback in the form of a reward. This reward serves as a direct indicator of the success or failure of the action relative to the task's objectives. For instance, successfully chopping vegetables into uniform pieces might yield a positive reward, while uneven or incomplete chopping would result in a lower reward. Task rewards are typically quantitative, providing a clear metric for evaluating performance.

Task rewards play a critical role in shaping the agent's decision-making process. By analyzing the rewards associated with different actions, the agent can identify patterns and correlations that inform its future choices. For instance, if the agent consistently receives higher rewards for using a particular skill or strategy, it might prioritize that approach in similar contexts. Over time, this feedback loop enables the agent to refine its behavior, optimizing its actions to achieve maximum rewards. Another important consideration is the temporal aspect of rewards. Some tasks produce immediate feedback, while others involve delayed rewards that depend on the completion of subsequent steps. For example, boiling water might not yield a reward until the water is used in cooking, creating a dependency on downstream actions. The agent must learn to associate delayed rewards with their initiating actions—a process that requires sophisticated temporal reasoning and memory.

The agent aims to maximize the task reward over time. This involves learning a policy that maps states to actions to interact with the environment such that the expected long-term reward is maximized. The reward function $r(\text{state}, \text{action})$ represents the intermediate reward of being in a state and taking an action. Denote policy π as a mapping from action space to state space. The goal is to find the optimal policy (π^*) that maximizes the cumulated reward, which can be formulated as the following problem:

$$\pi^* = \arg \max_{\pi} E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(\text{state}_t, \text{action}_t) \right] \quad (8)$$

where γ is the discount factor, and the expectation E is taken over the trajectory $\{\text{state}_t, \text{action}_t\}$ generated by the policy π . The agent updates its policy iteratively to maximize the cumulative value feedback, thereby aligning its actions with its long-term objectives.

Value feedback is inherently multidimensional, reflecting the complex interplay between different value dimensions. This creates both opportunities and challenges for the agent. On the one hand, value feedback allows the agent to evaluate its performance holistically, considering both immediate and long-term impacts. On the other hand, it requires the agent to navigate trade-offs between competing values, such as balancing health and efficiency or individual and collective benefits. Resolving these trade-offs involves sophisticated reasoning and prioritization, guided by the agent's cognitive model and selectors. Another challenge is the ambiguity and variability of value feedback. Unlike task rewards, which are often quantitative and objective, value feedback can be subjective and context-dependent. For example, the same action might be evaluated differently depending on the agent's current priorities or the perspectives of other stakeholders. The agent must learn to interpret and integrate these nuanced signals, developing a robust understanding of how its actions influence its value dimensions over time. Value feedback also plays a critical role in shaping the agent's long-term learning and adaptation. By analyzing trends and patterns in value feedback, the agent can identify opportunities for growth and improvement, refining its skills and strategies to better align with its objectives. For instance, if the agent consistently receives negative feedback for wasting resources, it might develop more efficient practices or prioritize sustainability in future tasks. This iterative process ensures that the agent remains aligned with its evolving goals and values.

It should be noted that the proposed framework is a unified learning paradigm that encompasses various existing machine learning paradigms, such as RL, passive learning, active learning, and learning from demonstrations. In particular, our framework addresses all three components of the (C, U, V) learning framework in the learning process of an intelligent agent. In contrast, other existing frameworks only cover one or two of these components. For instance, passive learning focuses solely on U , active learning integrates both U and V , and RL encompasses C . Therefore, existing frameworks can be regarded as a subset of our framework.

6. Learning problems in the (C, U, V) space: Completeness, convergence, and limits

In this section, we first discuss the completeness issue in AGI, including cognitive completeness and task completeness.

6.1. Cognitive completeness

With the (C, U, V) learning framework, our intent is to create a model that is powerful enough to simulate the thinking and decision-making process of any human. However, the human brain is complex, and different people will view the same scene differently. Thus, the aim of cognitive completeness is to understand

whether the parameters have the capacity to generate the representation of any given scene for any person. More specifically, for a particular scene x , we aim to establish a PG space equivalent to the cognitive space a human would have. In our (C, U, V) framework, the agent constructs a PG based on the current scene and the selected skills from the U space using the set of parameters θ_{PG} .

We call the (C, U, V) space cognitively complete if, for any PG in the cognitive space, there exists a set of parameters θ_{PG} that can generate it. If the (C, U, V) space defined by a specific function class (e.g., a deep neural network) is cognitively complete, and the generated PGs can express all reasonable human cognition and possible actions for the given scene, we denote the framework as possessing cognitive completeness for that scene, meaning that it is capable of capturing every aspect of the scene and all potential interactions, aligning closely with human understanding and capabilities.

Let us consider the example of the kitchen scene. In this scenario, reasonable human cognition and complex actions include tasks such as washing vegetables, chopping ingredients, cooking, and cleaning. These actions can be further broken down into simpler behaviors such as visual processing, movement, and the use of specific tools. In fact, the actions in this scene are arbitrary. For instance, a human might choose to dismantle the entire kitchen. However, we want the intelligent agent to understand the "reasonable" actions that humans take in this scenario rather than "all possible" actions and to build its cognitive completeness based on this understanding. If the agent's U space allows it to form PGs that encompass all reasonable actions, then we consider the agent to possess cognitive completeness for this scene. This means that the agent's cognitive architecture is capable of representing and understanding all plausible human behaviors and interactions within the kitchen environment.

6.2. Task completeness

Task completeness refers to the ability of an intelligent agent to achieve its goals through a series of actions that increase the values in its V -dimension. More specifically, an agent's actions unfold within the current PG and lead to the formation of new PGs through iteration. Thus, the agent has a future target PG. If an agent has the capability to find all possible target PGs that can increase the values in its V -dimension, as well as the capability to determine a path from the agent's current PG to the future target PG, allowing the agent to achieve its ultimate goal through a series of actions, then the agent satisfies task completeness. This means that the agent's planning and execution capabilities are sufficient to navigate from its current state to the desired state, ensuring that all necessary steps and actions are accounted for to reach the goal. Using the kitchen scenario as an example, the agent's future goal might be to learn how to prepare a delicious meal, while its current PG might only guide it to wash vegetables. If the agent's (C, U, V) framework includes sufficient cognitive and learning potential to allow it to gradually learn how to wash vegetables, chop ingredients, cook, and continuously improve the taste of its dishes from failures, then—regardless of how long the process takes—the agent will eventually be able to achieve the task of preparing a delicious meal. In this case, the agent possesses task completeness.

Next, we discuss the problem of learning limits for general intelligent agents in the (C, U, V) system. In the previous section, we introduced the (C, U, V) learning framework to describe the set of physical laws, social norms, and value functions mastered by an intelligent agent, which are used to characterize the agent's behavior. We also described an agent's learning process through the iteration and evolution of the (C, U, V) system. In this subsection, we draw an analogy to the convergence problem in computer science and introduce a similar research problem in agent learning.

For clarity, we say that the (C, U, V) system converges as the parameters in (C, U, V) converge. More specifically, we examine the conditions under which the learning process will terminate at different (C, U, V) equilibrium points and investigate the limit of (C, U, V) evolution. The answers to these questions determine the fundamental limits of learning. First, we must consider an individual intelligent agent and where it will tend to evolve within the (C, U, V) learning framework, much like the balance achieved in teaching and learning in our daily lives. Next, we must consider this in the context of a population, when treating all intelligent agents as a population. As each agent learns and evolves within its (C, U, V) framework, we consider how the entire population will ultimately evolve and what kind of equilibrium it will reach. This represents the limits that the entire intelligent community can achieve through learning.

6.3. Convergence of intelligent agents

Learning is the fundamental driving force behind the intelligence in agents. The learning process is essentially the evolution of the (C, U, V) system. As the (C, U, V) system of a single intelligent agent evolves, at what state will it ultimately reach a balance? It is clear that the evolutionary dynamic of the (C, U, V) system depends on the algorithm applied by the agent. Different algorithms lead to different dynamics and thus to different final states of the agents. A basic criterion to judge the algorithm is the convergence of the dynamic. We start with the following definition:

Definition 1. Denote (C_0, U_0, V_0) as the initialization of (C_t, U_t, V_t) at time $t = 0$. Consider any algorithm that drives the (C, U, V) system with this initialization. The algorithm is said to be convergent concerning U if there exists a limit point U_∞ such that

$$\lim_{t \rightarrow \infty} U_t = U_\infty \quad (9)$$

Similarly, the algorithm is said to be convergent concerning V if $\lim_{t \rightarrow \infty} V_t = V_\infty$, and the algorithm is said to be convergent concerning C if

$$\lim_{t \rightarrow \infty} C_t = C_\infty \quad (10)$$

A simple example of (C, U, V) convergence is the RL problem. Here, the value function is specified by a predetermined task reward and remains unchanged over time, indicating that the agent focuses exclusively on solving this specific task while disregarding everything else in the environment. Moreover, if feasible RL methods are applied to optimize the policy, such as a policy gradient, which provably maximizes the reward function, then the system will converge. Here, we take another simple case from nature to introduce the possible convergent scenarios of the evolution of a single agent's (C, U, V) system. Imagine a monkey starting from birth and undergoing a process of (C, U, V) system evolution through learning and interacting with its environment. The evolution of the monkey's C space represents the monkey's internal drive to choose values and learn skills. For a young monkey, this drive prioritizes learning basic behaviors such as crawling and eating. As the monkey matures, this internal drive shifts toward learning more advanced skills such as hunting and fighting. The evolution of the U space can be seen as the process of gradually learning and mastering basic survival skills, while the evolution of the V space refers to the changes that occur in the monkey's value functions at different stages of its life. By analyzing the upper limits and evolutionary boundaries of the monkey's (C, U, V) system evolution, we can more intuitively explain the three possible conditions under which an agent's (C, U, V) system may evolve and reach equilibrium.

6.3.1. U-space convergence

During the process of learning and growth, an intelligent agent may gradually understand the intentions of its surrounding instructors and the value of various strategies. However, due to limitations within the agent's own system, it may lack the capacity to develop further, ultimately leading to the convergence of the agent's learning. For instance, a monkey interacting with its environment or guided by the elders in its group can gradually acquire essential survival skills, such as climbing trees and gathering fruit. However, for an individual monkey, once it has mastered these necessary survival abilities, it may find it difficult to learn skills that exceed its cognitive or physical limits due to the limited development of its biological brain. Different individuals have different upper limits on their abilities. For example, fighting techniques may vary significantly between individuals, even among the same species or against other species. Stronger individuals may gradually master these skills and even become the alpha of the group, whereas weaker individuals might only reach their limit by acquiring basic survival skills. Additionally, a monkey cannot learn to fly, regardless of how beneficial this might be; a monkey simply lacks wings. In this case, we consider that the evolution of the U space for the individual monkey has reached its limit, leading to the convergence of learning (evolution).

6.3.2. V-space convergence

When an intelligent agent's value system is constrained by its own architecture, it may not recognize the purpose or benefits of continued learning, which can also lead to a convergence in learning. Taking the monkey as an example, most individual monkeys are likely to believe that valuable behaviors are those related to feeding, survival, and reproduction. Once they have mastered basic survival skills, they may perceive no other activities or skills as more valuable and therefore may stop striving for self-improvement, essentially "lying flat." However, a few individuals who have the tendency to compete for mates or territory may gradually learn combat skills and eventually challenge the alpha at the right time. In this scenario, we consider that differences in cognition or values determine the learning limit for an individual monkey, meaning that the evolution of the V space has reached its limit, resulting in convergence.

6.3.3. C-space convergence

The cognitive architecture of an intelligent agent is responsible for scheduling and selecting skills from the U space and value functions from the V space and then generating tasks that increase the selected values. When an agent's cognitive architecture determines that the agent's current skills have reached the upper limit of what the agent can master, or that the agent's current value functions have reached the optimal level the agent can achieve, the cognitive architecture cannot further orchestrate the learning process, leading to the convergence of learning. Taking the monkey as an example, when an individual has mastered all possible skills, including actions related to feeding, reproduction, and fighting, or when it has achieved all possible values, such as satiety, reproduction, and territory competition, its cognitive architecture cannot further select higher values or find new actions. This results in that individual monkey reaching its learning limit. At this point, the evolution of the monkey's C space has reached its limit, leading to convergence.

6.3.4. (C, U, V) space dynamics

During the learning process, the cognitive system, the skill set, and the values may gradually form a dynamic equilibrium. The (C, U, V) system can then tend toward a dynamic equilibrium state. To better formalize these dynamics, we use a continuous-time differential equation to describe the (C, U, V) evolution:

$$\frac{d(C, U, V)}{dt} = \text{Evo}(C, U, V, t) \quad (11)$$

where $\text{Evo}(C, U, V, t)$ is a vector field describing the dynamic relationship between (C, U, V) at time t ; the differential equation captures how (C, U, V) evolve over time; and the goal is to analyze the behavior of (C_t, U_t, V_t) that follows this equation. By concretely specifying the function $\text{Evo}(C, U, V, t)$, we can analyze the dynamics of the (C, U, V) system for specific problems. Below, we introduce several examples.

Case 1. (Ensuring convergence of a linear (C, U, V) system). When $\text{Evo}(C, U, V, t)$ is a linear function, the (C, U, V) system follows an autonomous linear dynamic. To be more concrete, we define the system as

$$\frac{d(C, U, V)}{dt} = \mathbf{A}[C, U, V] \quad (12)$$

where $\mathbf{A}$ is a symmetric dynamic matrix of the corresponding system. The solution to this dynamic system is known to be

$$[C_t, U_t, V_t] = e^{\mathbf{A}t}[C_0, U_0, V_0] \quad (13)$$

Denote $\mathbf{I}$ as the identity matrix. To investigate the convergence property of the system when t approaches infinity, we expand $e^{\mathbf{A}t}$ as

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2}{2!}t^2 + \dots \quad (14)$$

Because $\mathbf{A}$ is a symmetric dynamic matrix, we decompose it to $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$, where $\mathbf{Q}$ is an orthogonal matrix and $\mathbf{\Lambda}$ is a diagonal matrix. Inserting this decomposition into Eq. (14) yields

$$\begin{aligned} e^{\mathbf{A}t} &= \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2}{2!}t^2 + \dots = \mathbf{I} + \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T t + \frac{\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T}{2!}t^2 + \dots \\ &= \mathbf{Q} \left[\mathbf{I} + \mathbf{\Lambda}t + \frac{\mathbf{\Lambda}^2}{2!}t^2 + \dots \right] \mathbf{Q}^T = \mathbf{Q}e^{\mathbf{\Lambda}t}\mathbf{Q}^T \end{aligned} \quad (15)$$

It can be clearly seen that the asymptotic behavior of $e^{\mathbf{A}t}$ depends on the eigenvalues of $\mathbf{A}$ (i.e., the diagonal elements in $\mathbf{\Lambda}$). If all real values in $\mathbf{\Lambda}$ are negative, the system will converge. Conversely, when $\mathbf{\Lambda}$ contains non-negative real values, the system will diverge. If all real values of $\mathbf{\Lambda}$ are zero and $\mathbf{\Lambda}$ includes complex elements, the system will exhibit periodic behavior.

Case 2. (Estimating the limits of a (C, U, V) system at a given time T). When $\text{Evo}(C, U, V, t)$ represents the diffusion process on a graph from $t = 0$, it can be expressed as

$$\frac{d(C, U, V)}{dt} = -L[C, U, V] \quad (16)$$

where L refers to the weights in the diffusion process. This process can be depicted as adopting the Euler method for the K updates to obtain the approximation $[\hat{C}, \hat{U}, \hat{V}]$ at time T , as follows :

$$[\hat{C}, \hat{U}, \hat{V}] = \left(1 - \frac{T}{K}L\right)^K [C_0, U_0, V_0] \quad (17)$$

where K is the number of total updates.

Proposition 1. The approximation error between the graph aggregation's results $(\hat{C}, \hat{U}, \hat{V})$ and analytical results (C, U, V) at time T can be depicted as follows:

$$\|e_r^{(K)}\| \leq \frac{T\|L\| \cdot \| [C_0, U_0, V_0] \|}{2K} (e^{T\|L\|} - 1) \quad (18)$$

Proof: The graph aggregation can be considered as a general Euler forward scheme with a step size h ,

$$[\hat{C}^{(k+1)}, \hat{U}^{(k+1)}, \hat{V}^{(k+1)}] = [\hat{C}^{(k)}, \hat{U}^{(k)}, \hat{V}^{(k)}] - hL[\hat{C}^{(k)}, \hat{U}^{(k)}, \hat{V}^{(k)}] \quad (19)$$

where $h = T/K$, in which k is the index of intermediate update. The analytical results are

$$[C, U, V] = e^{-TL}[C_0, U_0, V_0] \quad (20)$$

We denote the error (e_k) between the analytical results and the Euler forward results at step k ($1 \leq k \leq K$) as follows:

$$e_k = [C^{(k)}, U^{(k)}, V^{(k)}] - [\hat{C}^{(k)}, \hat{U}^{(k)}, \hat{V}^{(k)}] \quad (21)$$

The truncation error $(TE^{(k)})$ of the Euler forward finite difference at step k is

$$TE^{(k)} = \frac{[C^{(k+1)}, U^{(k+1)}, V^{(k+1)}] - [C^{(k)}, U^{(k)}, V^{(k)}]}{h} + L[C^{(k)}, U^{(k)}, V^{(k)}] \quad (22)$$

which is equivalent to

$$[C^{(k+1)}, U^{(k+1)}, V^{(k+1)}] = [C^{(k)}, U^{(k)}, V^{(k)}] + h(TE^{(k)} - L[C^{(k)}, U^{(k)}, V^{(k)}]) \quad (23)$$

Taking the difference between the above equation and the Euler forward scheme's equation, we obtain

$$e_{k+1} = (1 - hL)e_k + hTE^{(k)} \quad (24)$$

whose norm can be upper bounded as

$$\|e_{k+1}\| \leq (1 + h\|L\|)\|e_k\| + h\|TE^{(k)}\| \quad (25)$$

If we let $M = \max_{0 \leq k \leq K-1} \|TE^{(k)}\|$ be the upper bound on global truncation error, we have

$$\|e_{k+1}\| \leq (1 + h\|L\|)\|e_k\| + hM \quad (26)$$

By induction, and noting that $1 + h\|L\| \leq e^{h\|L\|}$ and $e^{(0)} = [C^{(0)}, U^{(0)}, V^{(0)}] - [\hat{C}^{(0)}, \hat{U}^{(0)}, \hat{V}^{(0)}] = 0$, we have

$$\|e_k\| \leq \frac{M}{\|L\|} \left[(1 + h\|L\|)^k - 1 \right] \leq \frac{M}{\|L\|} (e^{Kh\|L\|} - 1) \quad (27)$$

From the Taylor theorem, we can obtain the truncation error $TE^{(k)}$ as follows:

$$TE^{(k)} = \frac{1}{2h} L^2 [C_\delta, U_\delta, V_\delta] \quad (28)$$

where $\delta \in [kh, (k+1)h]$, such that the truncation error $TE^{(k)}$ can be bounded by

$$\begin{aligned} \|TE^{(k)}\| &= \frac{1}{2h} \|L\|^2 \| [C_\delta, U_\delta, V_\delta] \| = \frac{1}{2h} \|L\|^2 \|e^{-\delta L}[C_0, U_0, V_0]\| \\ &\leq \frac{1}{2h} \|L\|^2 \| [C_0, U_0, V_0] \| \end{aligned} \quad (29)$$

Finally, when $k = K$, we have

$$\begin{aligned} \|e_K^{(T)}\| &\leq \frac{\|L\| \cdot \| [C_0, U_0, V_0] \|}{2h} (e^{Kh\|L\|} - 1) \\ &= \frac{T\|L\| \cdot \| [C_0, U_0, V_0] \|}{2K} (e^{T\|L\|} - 1) \end{aligned} \quad (30)$$

The proof is complete.

Case 3. $((C, U, V)$ system with a gradient field). Here, $\text{Evo}(C, U, V, t)$ represents an autonomous gradient field of a energy F , such that

$$\frac{d(C, U, V)}{dt} = -\nabla F(C, U, V) \quad (31)$$

Then (C, U, V) will evolve toward a minimum of F . The convergent properties of this system are closely tied to the property of F . We first provide a guarantee of the convergence of the (C, U, V) space

dynamics with a gradient field for general energy functions. More specifically, through the chain rule, the derivative of F with respect to t can be expressed as shown below.

Proposition 2. Consider F is continuously differentiable under the (C, U, V) system with a gradient field. Then $F(C, U, V)$ is monotonically decreasing as

$$\frac{dF}{dt} = -\|\nabla F(C_t, U_t, V_t)\|^2 \quad (32)$$

The conclusion is obvious. When the gradient ∇F is large at initialization, F decreases rapidly; when the gradient ∇F is small, F decreases slowly until convergence, when F is bounded below. This ensures the convergence of F , but the convergence of F does not necessarily imply the convergence of (C, U, V) . In some cases, the convergence of F is enough. However, for some cases, it is not sufficient, and further characterization of the convergence of (C, U, V) is necessary. Under the local Kurdyka–Łojasiewicz (KL) condition, the convergence properties of the (C, U, V) dynamics can be enhanced. More specifically, we can show that (C, U, V) will converge to a local optimal value.

Proposition 3. Assume F satisfies the local KL condition with the set of parameter $\theta \in [0, 1)$ in a neighborhood of certain (C^*, U^*, V^*) . Denote F^* as the global minima of F . There exists a constant $\eta > 0$ such that for all (C_t, U_t, V_t) with $0 < \|(C_t, U_t, V_t) - (C^*, U^*, V^*)\| \leq \eta$,

$$\|\nabla F(C_t, U_t, V_t)\| \geq (F(C_t, U_t, V_t) - F^*)^\theta \quad (33)$$

Then, the dynamic of $F(C_t, U_t, V_t)$ converges exponentially to F^* as follows:

$$C_t, U_t, V_t \rightarrow C^*, U^*, V^* \quad (34)$$

Next, we focus on the local properties of the solutions of the (C, U, V) space dynamics. Specifically, the evolution of the (C, U, V) framework is influenced by value feedback, which is inherently fuzzy and variable. This fuzziness and variability introduce a certain degree of jumpiness in the system's dynamics, such that the evolution process must deal with complex functional landscapes and realize the transitions of values and capabilities. To mathematically characterize and analyze this jumpiness, we can model it as a saddle escaping problem. Understanding these properties is crucial to ensure that the (C, U, V) dynamics converge to meaningful and stable solutions with general energy functions. To simplify, we denote $z = (C, U, V)$ to represent the parameter space and analyze how the (C, U, V) dynamics with a gradient field escape the saddle point.

Proposition 4. Let F , as defined above, be a twice continuously differentiable function. Let e be the eigenvector corresponding to the smallest eigenvalue of its Hessian matrix $\nabla^2 F(z_0)$ as $-\delta$, and let $q > 0$ be a small scalar. Consider two evolution processes—namely, $w_t(u)$ and a perturbation $w_t(u + qe)$ —generated by the (C, U, V) dynamics with the gradient field, initialized from a neighborhood point denoted by u with a small disturbance q along the direction of the smallest eigenvalue of $\nabla^2 F(z_0)$ denoted by e . Then, at least one of the processes $w_t(u)$ or $w_t(u + qe)$ will escape the neighborhood of z_0 as the evolution proceeds.

Proof: We consider the following two dynamics:

$$\frac{dw_t}{dt} = -\alpha \nabla F(w_t) \quad (35)$$

and

$$\frac{d\bar{w}_t}{dt} = -\alpha \nabla F(\bar{w}_t) \quad (36)$$

where $\bar{w}_t \equiv w_t(u + qe)$, and α stands for a positive step size. Then, we couple the difference process $\bar{w}_t - w_t$ and keep moving the iteration afterward as $z_t = \bar{w}_t - w_t$. By tracking the dynamic of z_t , we can show that, for any u and $u + qe$ in the neighborhood of x_0 , at least one of the evolution processes $w_t(u)$ or $w_t(u + qe)$ will escape the saddle point. First, we show the dynamic of z_t as follows:

$$\begin{aligned} \frac{dz_t}{dt} &= -\alpha(\nabla F(\bar{w}_t) - \nabla F(w_t)) \\ &= -\alpha \left[\int_0^1 \nabla^2 F(w_t + \theta(\bar{w}_t - w_t)) d\theta \right] z_t \\ &= -\alpha(H - D_t)z_t \end{aligned} \quad (37)$$

where $H = \nabla^2 F(x_0)$, and $D_t = \nabla^2 F(x_0) - \int_0^1 \nabla^2 F(w_t + \theta(\bar{w}_t - w_t)) d\theta$. It can be seen that D_t is upper bounded as follows, when assuming that F has a ρ -Lipschitz continuous Hessian.

$$\begin{aligned} \|D_t\| &\leq \int_0^1 \|\nabla^2 F(x_0) - \nabla^2 F(w_t + \theta(\bar{w}_t - w_t))\| d\theta \\ &\leq \rho \int_0^1 \|\theta(\bar{w}_t - x_0) + (1 - \theta)(w_t - x_0)\| d\theta \\ &\leq \rho \max(\|\bar{w}_t - x_0\|, \|w_t - x_0\|) \end{aligned} \quad (38)$$

Assume that all (C, U, V) states are globally bounded in the B -neighborhood of x_0 . We obtain $\|D_t\| \leq \rho B$. Moreover, we analyze the difference process z_t in two aspects: the rotation effect in the direction of e and the expansion effect of the norm. We introduce $\psi_t \equiv q^{-1}(1 + \alpha\delta)^{-t} z_t$. Then, we have

$$\frac{d\psi_t}{dt} = -(\alpha H + \ln(1 + \alpha\delta))\psi_t + \alpha D_t \psi_t \quad (39)$$

We first bound the norm of ψ_t . $\|\psi_0\| = \|e\| = 1 \leq 2$, supposing that $\|\psi_t\| \leq 2$ for all $t < T$. Then,

$$\begin{aligned} \|\psi_T\| &\leq 1 + \int_0^T \| -(\alpha H + \ln(1 + \alpha\delta))\psi_t + \alpha D_t \psi_t \| dt \\ &\leq 1 + 2(\alpha\rho + \alpha\delta)T + 2\alpha\rho BT \end{aligned} \quad (40)$$

By letting $T = \frac{1}{4\alpha \max(\rho, \delta)}$ and $B = \frac{\delta}{\rho}$, we obtain $\|\psi_T\| \leq 2$. This means that the norm of the differential process z_t grows exponentially with time t ; however, it can be controlled. We then consider the expansion effect as follows:

$$e^T \psi_t \geq 1 - \int_0^T \|\alpha D_t \psi_t\| dt \geq 1 - 2\alpha\rho BT \geq \frac{1}{2} \quad (41)$$

Eq. (41) indicates that, as the evolutionary time progresses, the distance of the two processes will grow in the negative curvature direction, ensuring that at least one of the two processes will escape from the saddle point if the difference norm is small.

The proof is complete.

6.4. Multi-agent (population) convergence

Due to differences in their environments and (C, U, V) spaces, different intelligent agents may reach different states during their learning processes. In the previous subsection, we discussed the learning process of a single intelligent agent. Next, we consider the collective learning of all agents, treating the entire population of agents as a whole and examining the evolutionary limits of the population. This is the problem of the intelligent learning limit, which involves the learning boundaries and ultimate level of intelligence of an entire intelligent community.

To study the learning limits of an entire intelligent population, we begin with traditional mean-field theory. In this approach, each intelligent agent is viewed as an instance within the population, and a distribution function is introduced to describe the (C, U, V) space of the entire population. Equations based on this distribution are used to characterize the evolution of the population over time, exploring the ultimate direction of the population's distribution. We provide the following definition against the convergence of the (C, U, V) system at the population level:

Definition 2. Consider algorithm Q that follows distribution P^Q . For any algorithm Q in the support set of P^Q , let U_t^Q and V_t^Q evolve by Q with the initialization (C_0, U_0, V_0) . Let P_t^C , P_t^U , and P_t^V be the distribution of C_t^Q , U_t^Q , and V_t^Q , respectively. Then, the (C, U, V) system is said to be convergent concerning U in the population if

$$\lim_{t \rightarrow \infty} P_t^U = P_\infty^U \quad (42)$$

Similarly, the system is said to be convergent concerning V in the population if $\lim_{t \rightarrow \infty} P_t^V = P_\infty^V$, and to be convergent concerning C in the population if

$$\lim_{t \rightarrow \infty} C_t^V = C_\infty^V \quad (43)$$

Here, convergence is described in a weak sense.

We use the case of a monkey population as an example to explain the learning limits problem for the entire intelligent community. The possible final states include the following:

- **U system convergence:** Over the course of learning, the entire population of agents may gradually stop learning due to insufficient abilities, such as low learning efficiency for a particular problem. For example, in a monkey population, even though individual abilities may vary (i.e., different monkeys have different U spaces), it is an obvious fact that monkeys can never learn to fly. This limitation is due to their physiological structure, representing a boundary of the U space for the entire population.
- **V system convergence:** The value system of the entire intelligent population may converge, leading to the belief that there is no need to continue learning, resulting in convergence. Similarly, for a monkey population that shares some similarities with humans, the population's U space would certainly include the physical ability to perform actions such as striking stone or wood together, which would eventually produce fire. However, the population's V space lacks the ability to understand the fact that these actions might lead to fire or to recognize the value of starting a fire by friction. Physiologically, the brain structure of monkeys is not as developed as that of humans, making it difficult for them to infer the value of fire-making. Therefore, even though their physiological structure might allow them to perform the action of rubbing sticks together, the monkey population does not recognize the value of this behavior. After mastering basic survival and reproduction skills, they do not continue to explore the principles governing the world, leading to V system convergence.
- **C system convergence:** The cognitive architecture of the entire intelligent agent population may also converge, preventing the agents from further discovering higher-level values and scheduling corresponding actions. Taking a monkey population as an example, the cognitive architecture of the monkey group has boundaries. This allows individuals to distinctly master different skills such as hunting and fighting, or to have varying perceptions of values such as reproduction and territory. However, no individual within the group can learn to fly or understand mathematical formulas. Therefore, the cognitive architecture of the entire species has its limits, representing the convergence of the C system.
- **(C, U, V) system dynamic equilibrium:** The intelligent population may form a dynamic equilibrium with respect to its environment. There is appropriate diversity and complementarity among the agents, allowing them to adapt to their environment through cooperation, thereby achieving a dynamic balance. In fact, the continued existence and reproduction of existing biological populations are due to their ability to achieve a dynamic equilibrium with their environment under the constraints of their (C, U, V) spaces. The monkey population, through all its acquired skills, continuously reproduces in nature, thus reaching a dynamic equilibrium with the environment.

We consider all intelligent agents as a population in order to explore the learning boundaries and cognitive limits of the entire intelligent community. Just as the ability limits of a monkey group lie in hunting, climbing, and fighting, but not in flying, the cognitive limits of a monkey group are focused on completing the tasks necessary for survival and reproduction but do not extend to further exploring the mysteries of the world. Based on the (C, U, V) system, we can study how large the ultimate (C, U, V) space of a general intelligent population can be under given conditions, where the population's ability and cognitive boundaries lie, and whether the cognitive space of the entire intelligent community is larger or has greater potential than the (C, U, V) space of humans. The learning limits and ultimate cognitive levels of the entire intelligent community are fundamental problems in the study of AGI.

7. Discussion

AGI is a hot topic that extends beyond computer science, attracting interest from diverse fields such as cognitive science, psychology, and philosophy. This interdisciplinary appeal stems from the potential AGI holds to revolutionize our understanding of intelligence, consciousness, and decision-making processes. Researchers aim to create systems capable of performing a broad range of intellectual tasks with human-like adaptability, raising profound questions about ethics, societal impact, and the future of human-machine coexistence. However, during this exploration, researchers from different fields have questions and concerns about the path toward AGI. In this section, we address these concerns and present our perspectives.

Question 1: What is the best path toward achieving AGI, and what constitutes good research in this exploration?

The best path toward achieving AGI remains an open and highly debated question, given the complexity and interdisciplinary nature of the field. While there is no single definitive answer, several principles and practices can guide this exploration toward meaningful progress. First of all, we believe AGI development requires insights from computer science, neuroscience, cognitive science, philosophy, psychology, and ethics. Each discipline contributes unique perspectives on understanding and replicating human-like intelligence. Fostering collaboration between disciplines ensures that AGI systems are built with both technical robustness and a deep understanding of human behavior and cognition. Second, good research should take safety and ethical alignment as its central requirements. This includes ensuring that AGI systems behave predictably, align with human values, and minimize the risk of unintended consequences. The last and most important principle is that the need to advance theoretical frameworks for intelligence—such as cognitive architectures, neural-symbolic systems, and novel learning paradigms—is critical because these foundations provide a basis for understanding generalization, reasoning, and adaptability. Research should explore not just how AGI can perform tasks but also why and how it arrives at its conclusions, thereby ensuring transparency and explainability.

From this perspective, we believe our approach holds great potential for advancing AGI research. By actively collaborating with experts from diverse fields, we aim to model the human cognitive system and develop a robust theoretical framework within this domain.

Question 2: Humans can give birth to other humans, but can an AGI create another AGI?

We believe that the issue of whether AGI agents can establish trust, self-confidence, and mutual trust is key in their self-discovery, self-replication, and continuous evolution. An agent that has reached the human level (C, U, V) is called a "Tong Agent." At present, the Tong Agent has a structured cognitive framework

but may lack the ability to fully understand its own structure without learning, akin to humans' limited comprehension of their own cognitive mechanisms. To address this, AGI must solve the problem of "trust" during its training process. Here, the concept of trust includes three dimensions: trust in others (A trusts B), mutual trust (A and B trust each other), and self-trust/self-awareness, encompassing multiple levels and dimensions such as self-awareness, consciousness, self-confidence, and self-esteem.

These issues are being explored in explainable AI projects, which are grounded in the theory of mind. Trust is divided into two layers: trust in *U* and trust in *V*. To achieve this, an AGI agent must understand itself, including its capabilities, skills, and values, in order to build multi-level, multidimensional self-awareness and self-trust. With this self-awareness, AGI can establish a theory of mind to recognize and build trust with others, as well as to understand itself, thus enabling collaboration. Ultimately, this foundation allows AGI agents to shape themselves and establish mutual trust and cooperation with other general intelligences, such as biological humans, digital agents, and robots.

Question 3: AGI is implemented in circuits, and its computational limits can be described by a Turing machine. Do we need an alternative framework to understand the limitations of AGI?

In fact, no matter how complex a program may be, its implementation on a computer is ultimately reduced to a long sequence of basic instructions in assembly language. However, when we are dealing with a complex program, we always choose to use high-level language, as it provides a more abstract representation that is closer to human thinking, making code easier to understand. As another example, the human body is composed of basic elements such as carbon, hydrogen, oxygen, and nitrogen. However, when examining the human brain or addressing disease treatment, no one starts by analyzing these fundamental elements; instead, we approach the problem from higher-level perspectives to understand its causes. Similarly, for the even more complex phenomenon of intelligence, we must employ higher-level abstractions, much like a "high-level language," to model humanity's intelligence from a more comprehensive and sophisticated viewpoint. This is why we seek a new framework to describe and understand such complexities.

Our framework depicts the construction of intelligent agents, which are composed of three elements: cognitive architecture, the skill library, and value system. These are supported by a cognitive framework, which enables the creation of agents with arbitrary complexity by integrating new value/skill systems. Although Turing machines, based on the Von Neumann architecture, can theoretically implement any agent through programming, ranging from artificial ants and bees to monkeys and even certain human functions, it is difficult to use a Turing machine to precisely describe the difference in different levels of intelligence. By introducing advanced abstractions in the form of the *C*, *U*, and *V* systems, our framework offers a more practical and innovative framework for describing and implementing complex intelligent agents, overcoming the limitations of Turing machines.

Question 4: Does AI require self-awareness and a value system? What is the ultimate goal of AGI? To act like human or to perform tasks that humans cannot?

We believe AI should be driven by a value system. Current models such as ChatGPT are trained using statistical learning, with the aim of mimicking human conversations. While this approach appears to have achieved significant success, showcasing the potential of AI without self-awareness, it also presents notable drawbacks. For instance, the training data is derived from a diverse range of human interactions, which inherently embed numerous—often conflicting—value systems. This can result in the AI generating responses with erroneous or even conflicting values that deviate from mainstream societal norms. We believe that value-driven

design is the right approach. By predefining clear values and principles to guide the behavior of intelligent agents, we can ensure the creation of AI systems that are not only effective but also practically valuable and aligned with societal needs.

Regarding the ultimate goal of AGI, we believe that AI must first emulate human behavior and understand human needs before it can effectively perform tasks that are beyond human capability. At its core, innovation is driven by the goal of addressing unmet human demands. For example, entrepreneurs develop products by identifying specific needs within certain groups that have not been adequately fulfilled. Only products that meet genuine needs can gain market acceptance. Similarly, intelligent agents must first develop a deep understanding of human needs in order to identify the right tasks and provide meaningful, impactful solutions.

CRedit authorship contribution statement

Di He: Writing – original draft, Methodology, Investigation. **Cong Fang:** Writing – original draft, Methodology, Investigation. **Yisen Wang:** Writing – original draft, Methodology, Investigation. **Yujia Peng:** Writing – review & editing, Visualization, Methodology. **Yizhou Wang:** Supervision, Conceptualization. **Song-Chun Zhu:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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